

Name: Evan Litzer
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1A.

Running the model with the default settings of batch = 16 and learning rate = 0.01 results in a testing accuracy of 48.27% and a maximum training accuracy of 71.74 % through 20 epochs. This means that the model is predicting at a much higher rate for predicting the 10 classes, which means it is learning from the features. The gap indicates that there is overfitting.

1B.

Batch	Test Accuracy	Train Accuracy	Analysis
8	34.43 %	41.4 %	Very small batch size, so gradient estimates noise and causes unstable training. Loss barely decreases and accuracy plateaus, since learning rate is too large for batch size.
16	48.27 %	71.7 %	The gap between training and testing accuracy shows that this is overfitting because it cannot learn patterns like CNN would.
32	52.59 %	85.59 %	There is now more of a stable gradient estimate, so the model can converge better. Training accuracy is increased marginally, while testing accuracy is up around 5% too. Alas, overfitting is still occurring.
64	54.07 %	90.14 %	Testing accuracy and training accuracy increase. The larger batch size updates the gradient with more accurate signal each step, yet there is overfitting from the large gap between training and testing.

1C.

Learning Rate	Test Accuracy	Train Accuracy	Analysis
0.01	48.27 %	71.74 %	The network shows that the model can learn features with default settings. Yet the gap between testing and training accuracy is still too large and shows that there is overfitting.
0.001	55.13 %	85.1 %	This is the best testing accuracy achieved across all of the runs. Loss steadily decreases as more epochs occur, which means the model still has not converged fully.
0.0001	52.13 %	54.79 %	The model learns very slowly and reaches a lower training accuracy than bigger learning rate values. There is less overfitting as there is a small gap in testing and training accuracy.
0.00001	34.07 %	33.61 %	The learning rate is too small in that it only reaches 34 %. The weight updates are so small that learning can barely take place.